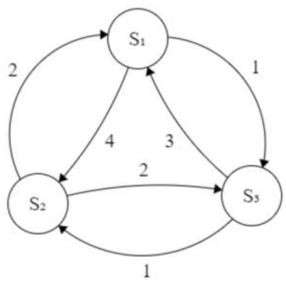
Lecture 21. Traveling Salesman Problem



$$\begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix}$$

Solution:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$x_{11} = 1$: The 1st stop is City #1
$x_{32} = 1$: The 2nd stop is City #3
$x_{23} = 1$: The 3rd stop is City #2

Setting up the energy function

Every column has exactly one city:

$$E_1 = (x_{11} + x_{21} + x_{31} - 1)^2 + (x_{12} + x_{22} + x_{32} - 1)^2 + (x_{13} + x_{23} + x_{33} - 1)^2$$

Every row has exactly one city:

$$E_2 = (x_{11} + x_{12} + x_{13} - 1)^2 + (x_{21} + x_{22} + x_{23} - 1)^2 + (x_{31} + x_{32} + x_{33} - 1)^2$$

Cost

$$L = \sum_{i,j,k}^{n} d_{ij}\, x_{ik} x_{j(k+1)}$$

$$L = \sum_{k=1}^{3} d_{12}\, x_{1k} x_{2(k+1)} + \sum_{k=1}^{3} d_{21}\, x_{2k} x_{1(k+1)} + \sum_{k=1}^{3} d_{13}\, x_{1k} x_{3(k+1)} + \sum_{k=1}^{3} d_{31}\, x_{3k} x_{1(k+1)}$$

$$+ \sum_{k=1}^{3} d_{23}\, x_{2k} x_{3(k+1)} + \sum_{k=1}^{3} d_{32}\, x_{3k} x_{2(k+1)}$$

$$L = \sum_{k=1}^{3} 4\, x_{1k} x_{2(k+1)} + \sum_{k=1}^{3} 2\, x_{2k} x_{1(k+1)} + \sum_{k=1}^{3} 1\, x_{1k} x_{3(k+1)} + \sum_{k=1}^{3} 3\, x_{3k} x_{1(k+1)}$$

$$+ \sum_{k=1}^{3} 2\, x_{2k} x_{3(k+1)} + \sum_{k=1}^{3} 1\, x_{3k} x_{2(k+1)}$$

$$L = \sum_{k=1}^{3} 4\, x_{1k}x_{2(k+1)} + \sum_{k=1}^{3} 2\, x_{2k}x_{1(k+1)} + \sum_{k=1}^{3} 1\, x_{1k}x_{3(k+1)} + \sum_{k=1}^{3} 3\, x_{3k}x_{1(k+1)}$$

$$+ \sum_{k=1}^{3} 2\, x_{2k}x_{3(k+1)} + \sum_{k=1}^{3} 1\, x_{3k}x_{2(k+1)}$$

$$\begin{aligned}
L = &\; 4x_{11}x_{22} + 4x_{12}x_{23} + 4x_{13}x_{21} \\
&+2x_{21}x_{12} + 2x_{22}x_{13} + 2x_{23}x_{11} \\
&+1x_{11}x_{32} + 1x_{12}x_{33} + 1x_{13}x_{31} \\
&+3x_{31}x_{12} + 3x_{32}x_{13} + 3x_{33}x_{11} \\
&+2x_{21}x_{32} + 2x_{22}x_{33} + 2x_{23}x_{31} \\
&+1x_{31}x_{22} + 1x_{32}x_{23} + 1x_{33}x_{21}
\end{aligned}$$

$$L = 4x_{22} + 2x_{23} + x_{32} + 3x_{33} + 2x_{22}x_{33+1x_{32}x_{23}}$$

$$E = L + \frac{\gamma}{2}E_1 + \frac{\gamma}{2}E_2$$

$$\begin{aligned}
E = &\; 4x_{11}x_{22} + 4x_{12}x_{23} + 4x_{13}x_{21} \\
&+2x_{21}x_{12} + 2x_{22}x_{13} + 2x_{23}x_{11} \\
&+1x_{11}x_{32} + 1x_{12}x_{33} + 1x_{13}x_{31} \\
&+3x_{31}x_{12} + 3x_{32}x_{13} + 3x_{33}x_{11} \\
&+2x_{21}x_{32} + 2x_{22}x_{33} + 2x_{23}x_{31} \\
&+1x_{31}x_{22} + 1x_{32}x_{23} + 1x_{33}x_{21} \\
&+\frac{\gamma}{2}(x_{11} + x_{21} + x_{31} - 1)^2 + \frac{\gamma}{2}(x_{12} + x_{22} + x_{32} - 1)^2 + \frac{\gamma}{2}(x_{13} + x_{23} + x_{33} - 1)^2 \\
&+\frac{\gamma}{2}(x_{11} + x_{12} + x_{13} - 1)^2 + \frac{\gamma}{2}(x_{21} + x_{22} + x_{23} - 1)^2 + \frac{\gamma}{2}(x_{31} + x_{32} + x_{33} - 1)^2
\end{aligned}$$

$$\begin{aligned}
E = &\; 4x_{22} + 2x_{23} + x_{32} + 3x_{33} + 2x_{22}x_{33} + 1x_{32}x_{23} \\
&+\frac{\gamma}{2}(x_{22} + x_{32} - 1)^2 + \frac{\gamma}{2}(x_{23} + x_{33} - 1)^2 \\
&+\frac{\gamma}{2}(x_{22} + x_{23} - 1)^2 + \frac{\gamma}{2}(x_{32} + x_{33} - 1)^2
\end{aligned}$$

You have 4 binary variables.
You need to evaluate 4 partial derivatives, to get 4 update equations.

You randomly or sequentially update the variables.

For example, you are considering x_{22},

$$\frac{\partial E}{\partial x_{22}} = 4 + 2x_{33} + \gamma(x_{22} + x_{32} - 1) + \gamma(x_{22} + x_{23} - 1) = 4 - 2\gamma$$

Initially all variables are set to 0. If $4 - 2\gamma > 0$, $x_{22} = 0$; if $4 - 2\gamma > 0$, then $x_{22} = 1$ at this step.

If $\frac{\partial E}{\partial x_{i_0 k_0}} > 0$, set $x_{i_0 k_0} = 0$.
If $\frac{\partial E}{\partial x_{i_0 k_0}} < 0$, set $x_{i_0 k_0} = 1$.